Yakeen Plus (2025)

Practice Test - 18

DURATION : 200 Minutes | **DATE : 26/12/2024** | **M. MARKS : 720**

Topics Covered

Physics :	Physics And Measurement, Kinematics, Laws Of Motion, Work, Energy, and Power, Rotational Motion
Chemistry :	Some Basic Concept of Chemistry, State of Matter, Solutions, Organic Chemistry: Some Basic principles and Techniques, (IUPAC Naming), Organic Chemistry: Some Basic principles and Techniques, (Isomerism), Organic Chemistry: Some Basic principles and Techniques, (GOC), Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure
Biology :	**(Botany):** Cell - The Unit of Life, Cell Cycle and Cell Division, The Living World, Biological Classification, Plant Kingdom **(Zoology):** Structural Organization in Animals, Animal Kingdom, Biomolecules, Breathing and Exchange of Gases

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min**. duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are **four Section** in the Question Paper, **Section I, II, III & IV** consisting of Section-I (Physics from question number **1 to 50**), Section-II (Chemistry: from question number **51 to 100**), Section-III (Botany: from question number **101 to 150**) & Section IV (Zoology: from question number **151 to 200**), having 50 Questions in each Subject and each subject is divided in two Section, Section A consisting 35 questions (all questions are compulsory) and Section B consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

NEET

Answer Key

Q1	(4)
Q2	(2)
Q3	(3)
Q4	(3)
Q5	(3)
Q6	(1)
Q7	(4)
Q8	(2)
Q9	(1)
Q10	(4)
Q11	(3)
Q12	(1)
Q13	(3)
Q14	(3)
Q15	(1)
Q16	(3)
Q17	(2)
Q18	(1)
Q19	(1)
Q20	(1)
Q21	(1)
Q22	(3)
Q23	(3)
Q24	(4)
Q25	(2)
Q26	(1)
Q27	(3)
Q28	(2)
Q29	(1)
Q30	(4)
Q31	(3)
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Q195 (4)

Q196 (4)

Q197 (1)

Q198 (3)

Q199 (4)

Q200 (4)

Hints & Solutions

Q1 Text Solution:

(4)

Let initial vel = u and

$v^2 = u^2 - 2gh \Rightarrow u^2 = 10^2 + 2 \times 10 \times 15$

$\Rightarrow u = 20 \text{ m/s}$

At $t = 3$ sec, $v = u - gt$

$\Rightarrow v = 20 - 3 \times 10 = -10 \text{ m/s}$

(i.e. 10 m/s down)

Q2 Text Solution:

(2)

$v = b\sqrt{x}$

$\frac{dv}{dt} = \frac{b}{2\sqrt{x}} \frac{dx}{dt} \quad a = \frac{bv}{2\sqrt{x}}$

$a = \frac{b(b\sqrt{x})}{2\sqrt{x}}$

$\frac{dv}{dt} = a = \frac{b^2}{2}$

$v = \frac{b^2}{2}\tau \quad \left[\text{At } t = \tau\right]$

Q3 Text Solution:

(3)

Let h be the height of the tower. Taking downward direction as positive, we get for the body projected upward,

$h = -ut_1 + \frac{1}{2}gt_1^2$ ......(1)

for the body projected downward,

$h = ut_2 + \frac{1}{2}gt_2^2$ ......(2)

for the freely falling body,

$h = 0 + \frac{1}{2}gt^2 = \frac{1}{2}gt^2$ ...... (3)

Multiplying equation (1) by t_2, we get

$ht_2 = -ut_1t_2 + \frac{1}{2}gt_1^2t_2$

Multiplying equation (2) by t_1, we get

$ht_1 = -ut_1t_2 + \frac{1}{2}gt_2^2t_1$

Adding (1) and (2), we get

$h(t_1 + t_2) = \frac{1}{2}gt_1^2t_2 + \frac{1}{2}gt_2^2t_1$

$= \frac{1}{2}gt_1t_2(t_1 + t_2)$

$\Rightarrow h = \frac{1}{2}gt_1t_2$

$\Rightarrow \frac{1}{2}gt^2 = \frac{1}{2}gt_1t_2$

$\{\because h = \frac{1}{2}gt^2 \text{from equation (3)}\}$

$\Rightarrow t^2 = t_1t_2$

$\Rightarrow t = \sqrt{t_1t_2}$

Q4 Text Solution:

(3)

Q5 Text Solution:

(3)

If displacement is linear function of time then velocity is constant that is motion will be of uniform nature.

Q6 Text Solution:

(1)

$\frac{dt}{dx} = (2\alpha x + \beta)$

$\therefore \frac{dx}{dt} = v = \left(\frac{1}{2\alpha x+\beta}\right)$

$a = \frac{dv}{dt} = -2\alpha\left(\frac{1}{2\alpha x+\beta}\right)^2 \cdot \frac{dx}{dt}$

$= -2\alpha(v^2)(v) = -2\alpha v^3$

Q7 Text Solution:

(4)

$a = \frac{dv}{dt}$

or $\boxed{\tan\theta = a}$

at point O particle in motion with constant speed.

Q8 Text Solution:

(2)

Initial velocity = $(\hat{i} + 2\hat{j})$ m/s

Magnitude of initial velocity,

$u = \sqrt{(1)^2 + (2)^2} = \sqrt{5}m/s$

Equation of trajectory of projectile is

$y = x\tan\theta$

$-\frac{gx^2}{2u^2}(1 + \tan^2\theta)\left[\tan\theta = \frac{y}{x} = \frac{2}{1} = 2\right]$

$\therefore \quad y = x \times 2 - \frac{10(x)^2}{2(\sqrt{5})^2}\left[1 + (2)^2\right]$

$= 2x - \frac{10(x^2)}{2\times5}(1 + 4)$

$= 2x - 5x^2$

Q9 Text Solution:

(1)

$a = 2v^2 \Rightarrow \frac{dv}{dt} = 2v^2$ or $\frac{dv}{dx} \times \frac{dx}{dt} = 2v^2$

$v\frac{dv}{dx} = 2v^2 \Rightarrow \frac{dv}{dx} = 2v$

$\int_{v_0}^{v} \frac{dv}{v} = \int_0^v 2dx \Rightarrow [\ln v]_{v_0}^{v} = [2x]_0^x$

$\ln\frac{v}{v_0} = 2x \Rightarrow v = v_0e^{2x}$

Q10 Text Solution:

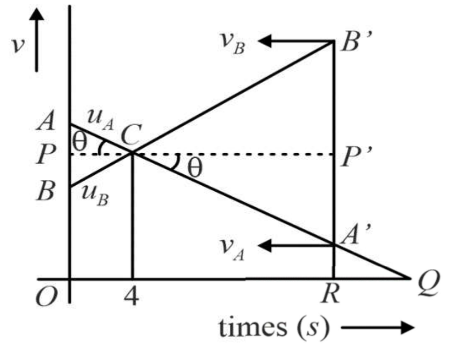
(4)

Firstly, the bolt is under inertial effect and then it falls freely.

Q11 Text Solution:

(3)

Let the particles meet at time t i.e., displacement of the particles are equal and which is possible when



Area of ΔACB = Area of $\Delta A'B'C$

[Area $OBCA'R$ being common]

$\Rightarrow$ Area of ΔAPC = Area $A'P'C$

$\frac{1}{2}AP \times PC = \frac{1}{2}A'P' \times P'C$

or $AP \times PC = A'P' \times P'C$

or $(PC \tan\theta)\ PC = (P'C \tan\theta)\ P'C$

or $PC = P'C$

$\therefore OR = PC + P'C = 2PC = 8s$

Q12 Text Solution:

(1)

$x = \frac{1}{t+5} \quad \therefore v = \frac{dx}{dt} = \frac{-1}{(t+5)^2}$

$\therefore\ a = \frac{d^2x}{dt^2} = \frac{2}{(t+5)^3} = 2x^3$

Now $\frac{1}{(t+5)} \propto v^{\frac{1}{2}} \quad \therefore \frac{1}{(t+5)^3} \propto v^{\frac{3}{2}} \propto a$

Q13 Text Solution:

(3)

$T = \sqrt{\frac{2h}{g}}$

$R = v.t$

$= v\sqrt{\frac{2h}{g}}$

Q14 Text Solution:

(3)

$U_x = \frac{dx}{dt} = 1$

and $U_y = \frac{dx}{dt} = 1 - 2t$

$\therefore\ U_{t=0} = \sqrt{u_x^2 + y_y^2} = \sqrt{1^2 + 1^2}$

$= \sqrt{2}\,\mathrm{m/s}.$

$a_x = \frac{d^2x}{dt^2} = 0$

$a_y = \frac{d^2y}{dt^2} = -2$

For time of flight,

$y = 0$

or $0 = t - t^2$

$\therefore t = 1s$

For maximum height, $t = \frac{1}{2}s.$

$\therefore\quad H = t - t^2 = \frac{1}{2} - \left(\frac{1}{2}\right)^2 = \frac{1}{4}\mathrm{m}.$

Q15 Text Solution:

(1)

Q16 Text Solution:

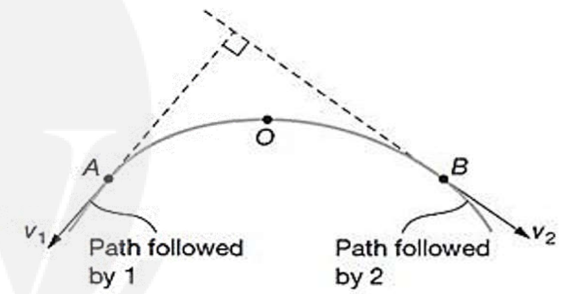
(3)

Let the velocities of the particles be at right angles to each other at time t. Then,

$\vec{v}_1 = \vec{u}_1 + \vec{g}t$

$\Rightarrow \vec{v}_1 = 3\hat{i} + gt\hat{j},\ \vec{v}_2 = \vec{u}_2 + \vec{g}t$

$\Rightarrow \vec{v}_2 = -4\hat{i} + gt\hat{j}$



For velocities to be mutually perpendicular, we have

$\vec{v}_1 . \vec{v}_2 = 0$

$\Rightarrow -12 + g^2t^2 = 0$

$t = \sqrt{\frac{12}{g^2}} = \frac{2\sqrt{3}}{g}s$

$x = (u_1 + u_2)t = 7\frac{2\sqrt{3}}{g}$

$\Rightarrow x = \frac{14\sqrt{3}}{g}$ m

Q17 Text Solution:

(2)

$x = 2t^3 + 4t \to V_x = 6t^2 + 4 \to a_x = 12t$

$y = 3t^2 + 7 \to V_y = 6t \to a_y = 6$

$t = 1\text{ sec}\ \vec{v} = 10\hat{i} + 6\hat{j},\ \vec{a} = 12\hat{i} + 6\hat{j}$

$v = \sqrt{100 + 36}\ ,\quad a = \sqrt{144 + 36}$

$= \sqrt{136}\ \mathrm{m/s} = \sqrt{180}\ \mathrm{m/s^2}$

Q18 Text Solution:

(1)

Q19 Text Solution:

(1)

$v = at + \frac{b}{t+c}$

Dimensionally, at = $v \Rightarrow a[T] = [LT^{-1}]$

$\Rightarrow a = [LT^{-2}]$

$c = t \Rightarrow c = [M^0L^0T^1]$

$\frac{b}{t+c} = v \Rightarrow \frac{b}{T} = [LT^{-1}] \Rightarrow [b] = [M^0LT^0]$

Q20 Text Solution:

(1)

$P = P_0 e^{-\alpha t^2+\beta t+\gamma}$

Exponent of e is dimensionless quantity.

$$-\alpha t^2 + \beta t + \gamma = \left(M^0L^0T^0\right)$$

$$\alpha t^2 = M^0L^0T^0$$

$$\alpha = \frac{M^0L^0T^0}{T^2}, \boxed{\alpha = M^0L^0T^{-2}}$$

$$\beta t = M^0L^0T^0$$

$$\beta = \frac{M^0L^0T^0}{T}, \boxed{\beta = M^0L^0T^{-1}}$$

$$\boxed{\gamma = M^0L^0T^0}$$

Q21 Text Solution:

(1)

$\text{nL.C.} = 0.01 \text{ mm}$

$\text{Total reading (T.R.)} = \text{MSD} + \text{CSR}$

$= (3 \text{ mm}) + 25\times(\text{L.C.})$

$= (3 \text{ mm}) + 25\times(0.01 \text{ mm}) = 3.25 \text{ mm}$

Q22 Text Solution:

(3)

$F = \mu N$

Q23 Text Solution:

(3)

$x^2 = ay$

$2x = a\frac{dy}{dx}$

$\frac{dy}{dx} = \frac{2x}{a}$

$\tan\theta = \frac{2x}{a}$

$\tan\theta \le \mu$

$\frac{2x}{a} \le \mu$

$$\boxed{x \le \frac{\mu a}{2}}$$

$y = \frac{1}{a} \times \frac{\mu^2 a^2}{4}$

$$\boxed{y = \frac{\mu^2 a}{4}}$$

Q24 Text Solution:

(4)

As block A does not slide on block B this means both blocks have the same acceleration along horizontal.

Net force on the system is $2F - F = F$

Therefore, Acceleration is $F/(m+m) = F/2m$.

Now taking of individual blocks. Let Normal between block be N.

Block on right:

Along Horizontal: $2F - N\cos 60 = ma$

$\Rightarrow 2F - m(F/2m) = N(1/2)$

$\Rightarrow \frac{3}{2}F = \frac{1}{2}N$

$\Rightarrow N = 3F$

Just to check we can calculate N at block on right also,

$N\cos 60 - F = ma$

$F + m(F/2m) = N(1/2)$

$\Rightarrow \frac{3}{2}F = \frac{1}{2}N$

$\Rightarrow N = 3F$

Q25 Text Solution:

(2)

Force at the surface of B and C

$N = 2m(a),\ a = \text{acceleration of system}$

$\therefore\ N = 2m\frac{F}{5m} = \frac{2F}{5}$

To prevent slipping of block B,

$\mu N = mg$

$\Rightarrow \frac{\mu 2F}{5} = mg,\ F = \frac{5}{2\mu}mg$

Q26 Text Solution:

(1)

$F(x) = -\frac{k}{2x^2}$ (Given)

$\Rightarrow mvdv = -\frac{k}{2x^2}dx$

$\Rightarrow mv\frac{dv}{dx} = -\frac{k}{2x^2}$ .... (1)

Integrating (1) within suitable limits

$\Rightarrow m\int_0^v vdv = -\frac{k}{2}\int_{1.0}^{0.5} x^{-2}dx$

$\Rightarrow \frac{1}{2}mv^2 = \frac{k}{2}\left[\frac{1}{0.5} - 1\right]$

$\Rightarrow v^2 = +1$

$\Rightarrow v = \pm 1 \text{ ms}^{-1}$

Since the particle is going from $x = 1$ m to $x = 0.5$ m, hence, its velocity is directed along the negative x-axis.

$\Rightarrow \vec{v} = (-1)\hat{i}$

Q27 Text Solution:

(3)

Let T be the tension in the rope and a and acceleration of rope.

The absolute acceleration of man is, therefore,

$\left(\frac{5g}{4} - a\right)$

Equations of motion for mass and man gives:

$T - 100g = 100a$

$T - 60g = 60\left(\frac{5g}{4} - a\right)$

Solving equation (i) and (ii), we get $T = \frac{4875}{4}\mathrm{N}$

Q28 Text Solution:

(2)

$\boxed{F = \frac{mv^2}{r}}$

Q29 Text Solution:

(1)

$\frac{dF}{dx} \Rightarrow (-) \Rightarrow \frac{dU}{dx}$ is also (–)

Hence particle will have stable equilibrium.

Q30 Text Solution:

(4)

$mg(h_1 - h_2) = \frac{1}{2}mV^2$

$V = \sqrt{2g(h_1 - h_2)} = \sqrt{2g(5.2m - 2m)}$

$V = 8\ \mathrm{m/s}\ H_1 = \frac{(V\sin\theta)^2}{2g}$

$\Rightarrow\ H_1 = \frac{(8\sin 45^\circ)^2}{2\times 10\ \mathrm{m/s}^2} = 1.6\ \mathrm{m}$

$\Rightarrow\ H_2 = \frac{(8\sin 30^\circ)^2}{2\times 10\ \mathrm{m/s}^2} = 0.8\ \mathrm{m}$

$\Delta H = H_1 - H_2 = 0.8\ \mathrm{m}$

Q31 Text Solution:

(3)

$W = \int \vec{F}\,.\,\vec{dr}$

Q32 Text Solution:

(1)

$e = \frac{\text{Velocity of separation}}{\text{Velocity of approach}}$

Q33 Text Solution:

(4)

$\boxed{F = -\frac{dv}{dx}}$

Q34 Text Solution:

(2)

Centripetal acceleration $a_c = k^2rt^2$

$\frac{v^2}{r} = k^2rt^2$

$\therefore v = Krt$

$\text{Now} \Rightarrow a_t = \frac{dv}{dt} = kr$

$\text{Power } P = f_t v = m\cdot a_t\cdot v$

$= mk^2r^2t$

Q35 Text Solution:

(1)

$W_{\text{total}} = \Delta KE$

$(3mg)x + mgx = \frac{1}{2}kx^2$

$\Rightarrow\ x = \frac{8mg}{k}$

Q36 Text Solution:

(1)

The ball rotates in a circle around the nail after is goes past the point *B*.

At point *B*, velocity of the ball is $v = \sqrt{2gL}$ ... (1)

Since the ball just completes the vertical circle about the nail, its speed at the lowest point should be

$v = \sqrt{5gr} = \sqrt{5g(L-d)}$ ...(2)

Equating (1) and (2), we get

$\sqrt{2gL} = \sqrt{5g(L-d)}$

$\Rightarrow 2L = 5L - 5d$

$\Rightarrow d = 0.6L$

Q37 Text Solution:

(1)

$E = \mathrm{KE} + V$

and $F = \frac{dv}{dx}$

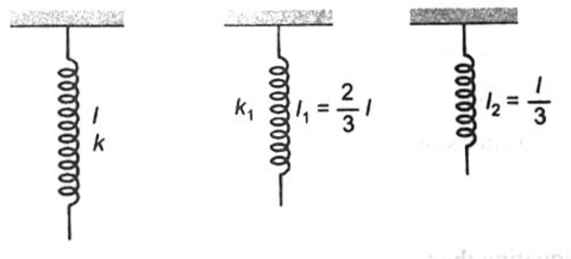
Q38 Text Solution:

(2)

$l_1 = 2l_2$

$\therefore\ l_1 = \frac{2}{3}l$

Force constant $l \propto \frac{1}{length\ of\ spring}$



$\therefore\ k_1 = \frac{3}{2}k$

Q39 Text Solution:

(2)

Nature of force at point *Q* restoring.

Q40 Text Solution:

(2)
By conservation of momentum.
$$\boxed{\vec{p_i} = \vec{p_f}}$$

Q41 Text Solution:
(3)
$$\boxed{\vec{p_i} = \vec{p_f}}$$
$$2mv = 3mv'$$
$$v' = \frac{2v}{3}$$

Q42 Text Solution:
(3)
$$p_i = p_f$$
$$mu = 2mv$$
$$v = \frac{u}{2}$$
$$\frac{1}{2}2m\left(\frac{u}{2}\right)^2 = 2mgh$$
$$\boxed{h = \frac{u^2}{8g}}$$

Q43 Text Solution:
(2)
The physics of this problems is that the center of mass (CM) cannot move. All of the force are "internal forces" so the CM of the system must remain in the same place. We can not wire the position of the CM before the switch and after the exchange, and then set them equal.
The CM from the center of plank is
(1) $x_{cm} = \frac{m_A x_A + m_B x_B + m_C x_C}{m_A + m_B + m_C + M}$
(2) $x_{cm1} = \frac{40.(-2)+50.0+60.2}{40+50+60+90} = \frac{1}{6}m$
$x_{cm2} = \frac{60.(-2)+50.0+40.2}{40+50+60+90} = -\frac{1}{6}m$

Thus, mass B will shift $x_{cm2} = \frac{2}{6} = \frac{1}{3}$m towards right 1/3 m towards right

Q44 Text Solution:
(1)
$$x_{\text{COM}} = \frac{(\sigma\pi^2 a)a - \sigma\pi\left(\frac{a}{2}\right)^2 \cdot \left(\frac{3a}{2}\right)}{\sigma\pi^2 a - \sigma\pi\left(\frac{a}{2}\right)^2} = \frac{a - \frac{3a}{8}}{1 - \frac{1}{4}} = \frac{5a}{6}$$

Q45 Text Solution:
(2)
At the time of maximum speed of point P, the rod should be vertical $\Delta K + \Delta U = 0$
$$\left[\frac{1}{2}\left(\frac{ml^2}{3}\right)\omega^2 - 0\right] + \left[-mg\frac{l}{2}\right] = 0$$
$$\Rightarrow \omega = \sqrt{\frac{3g}{l}}$$
Velocity of point P, $v_P = \omega l = \sqrt{3gl}$
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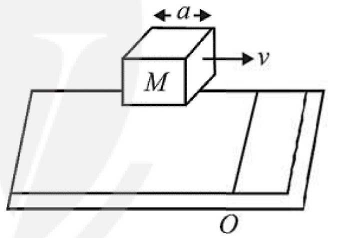
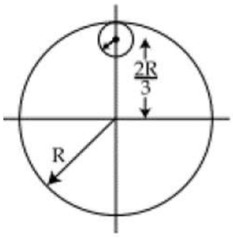
Q46 Text Solution:
(4)
Since, $E_K = \frac{L^2}{2I}$ (where I is M.I.)
$$\log E_K = \log L^2 - \log(2I)$$
$$\log E_K = 2\log L - \log(2I)$$
$\because$ I is constant.
Hence, above equation may be compared to $x = 0$ $y = mx - c$, hence intercept on y axis leads to graph (4)

Q47 Text Solution:
(1)
Angular momentum of block w.r.t O before collision at $O = mv\frac{a}{2}$. One collision, the block will rotate about the side passing through O. Now its angular momentum = $I\omega$.



By law of conservation of angular momentum,
$$Mv\frac{a}{2} = I\omega$$
$$\Rightarrow Mv\frac{a}{2} = \left(\frac{Ma^2}{6} + \frac{Ma^2}{2}\right)\omega$$
$$\Rightarrow \omega = \frac{3v}{4a}$$
Where I is moment of inertia of the block about the axis perpendicular to the plane passing through O.

Q48 Text Solution:
(2)
Given: From a uniform circular disc of radius R and mass $9M$, a small disc of radius $\frac{R}{3}$ is removed as shown in the figure.



To Find: The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through centre of disc.

Mass ∝ Area, $M \propto R^2$ Mass of portion removed

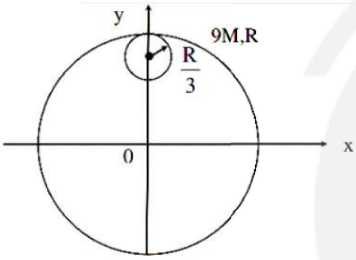
$\frac{M_1}{M_2} = \frac{1}{9}, M_1 = \frac{M_0}{9} = M$

$I_0 = \frac{9MR^2}{2} - \left[\frac{M\left(\frac{R}{3}\right)^2}{2} + M\left(\frac{2R}{3}\right)^2\right]$

$I_0 = \frac{9MR^2}{2} - \left[\frac{MR^2}{2} + \frac{4MR^2}{9}\right]$

$I_0 = \frac{9MR^2}{2} - \left[\frac{9MR^2}{18}\right] = \frac{9MR^2}{2} - \frac{MR^2}{2}$

$I_0 = 4MR^2$



Hence, The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through centre of disc is 4 MR^2.

Q49 Text Solution:

(2)

Moment of inertia of a uniform rod about one end $= \frac{mL^2}{3}$

∴ Moment of inertia of the system, which rod is bent

$= 2 \times \left(\frac{M}{2}\right)\frac{(L/2)^2}{3} = \frac{ML^2}{12}$

Q50 Text Solution:

(4)

Taking torque about the attachment point for W, we get

Taking torque about the attachment point for W, we get

$-T_1$ (0.4 L) + T_2 (0.3 L) + 500 (0.2 L) = 0

T = 1000 N, where $T_1 = T_2 = T$

$\Sigma F_y = 0$

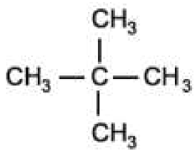
⇒ 2T – W – 500 = 0

⇒ W = 1500 N.

Q51 Text Solution:

(4)

Neopentane:



Q52 Text Solution:

(2)

$\%$ ionic character $= 16\left(\chi_A - \chi_B\right) + 3$
$.5\left(\chi_A - \chi_B\right)^2$

= 16(3 – 2) + 3.5 (3 – 2)^2 = 16 + 3.5 = 19.5%

Covalent character = 100 – 19.5 = 80.5%

Q53 Text Solution:

(4)

Group number for d-block = No. of electrons in (n – 1)d subshell + number of electrons in valence shell (nth shell)

∴ Group number of X = 5 + 1 = 6

Q54 Text Solution:

(1)

Valency of the element $= \frac{2\times \text{ vapour density}}{\text{equivalent weight } +35.5}$

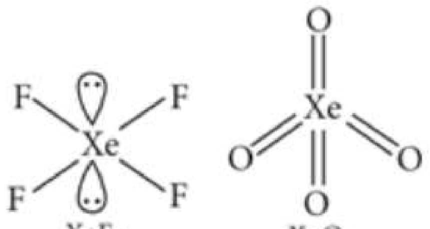
$= \frac{2\times 59.25}{9+35.5} = 2.662$

Atomic weight of the element = 9 × 2.662 = 23.96 u

Q55 Text Solution:

(3)

XeF_2 and I_3^- are sp^{3}d hybridised and are linear and hence, both are isostructural. NH_3 and PH_3, both are pyramidal and central atom in both cases is sp^3 hybrididsed. $SiCl_4$ and PCl_4^+, both are tetrahedral and central atom in both cases is sp^3 hybridised.



XeF_4 XeO_4

In XeF_4, Xe is sp^3d^2 hybridised and structure is square planar while in XeO_4, Xe is sp^3 hybridised and structure is tetrahedral.

Q56 Text Solution:

(1)

Amongst isoelectronic species, smaller the positive charge, smaller is the ionic radius is the wrong statement. Actually, it should be smaller the positive charge, bigger the ionic radius.

Q57 Text Solution:

(4)

Molarity = $\frac{0.967/159.5}{20/1000} = 0.303\ \mathrm{M}$

Q58 Text Solution:

(3)

1 mol of NH_3 requires $\frac{5}{4}$=1.25 mol of O_2 for complete reaction.

Hence, O_2 is the limited amount and will be completely consumed in the reaction, producing 4/5 mol of NO and 6/5 mol of H_2O.

Q59 Text Solution:

(3)

Molecular weight of AB_2 = $\frac{16}{0.25} = 64$

Molecular weight of AB_3 = $\frac{20}{0.25} = 80$

Let atomic weight of A and B be x and y respectively.

Then, x + 2y = 64

x + 3y = 80 or x = 32 and y = 16 g/mol

Q60 Text Solution:

(2)

α–Hydrogen atom with respect to C = C bond takes part in Hperconjugation

Q61 Text Solution:

(1)

Both $K_4[Fe(CN)_6]$ and $Fe_2(SO_4)_3$ dissociates to give 5 ions.

Q62 Text Solution:

(4)

16 g O_2 = 0.5 mol, 3 g H_2 = 1.5 mol,

Total number of moles, n = 2

$\mathrm{V} = \frac{\mathrm{nRT}}{\mathrm{P}} = \frac{2 \times 0.0821 \times 273}{1} = 44.8\ \mathrm{L} = 44800\ \mathrm{mL}$

Q63 Text Solution:

(4)

Priority order:

$$-\mathrm{OH} > -\mathrm{NH_2} > -\overset{|}{\mathrm{C}} = \overset{|}{\mathrm{C}}- > -\mathrm{C} \equiv \mathrm{C} -$$

Q64 Text Solution:

(2)

$\mathrm{P_{total}} = \mathrm{p_A} + \mathrm{p_B} = x_\mathrm{A} \mathrm{p_A^\circ} + x_\mathrm{B} \mathrm{p_B^\circ}$

$184 = \frac{3}{5} \times 200 + \frac{2}{5} \times \mathrm{p^\circ{}_B}$

$184 = 120 + \frac{2}{5}\ \mathrm{p^\circ{}_B}$

$\frac{5}{2}\mathrm{p_B^\circ}$ = (184-120)

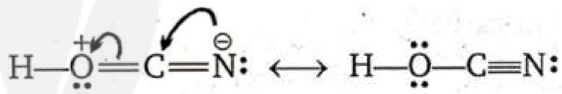
$\mathrm{p_B^\circ} = 160\ \mathrm{torr}.$

vapour pressure of pure liquid B is 160 torr.

.

Q65 Text Solution:

(2)



are resonationg structures.

Q66 Text Solution:

(3)

Let w g of each SO_2, CH_4 and O_2 are mixed

mole fraction of $CH_4 = \frac{n_{CH_4}}{n_{SO_4} + n_{CH_4} + n_{O_2}}$

$= \frac{\frac{w}{16}}{\frac{w}{64} + \frac{w}{16} + \frac{w}{32}} = \frac{4}{7}$

(Molecular weight of SO_2 = 64 u, CH_4 = 16 u, O_2 = 32 u)

p_{CH_4} = Mole fraction × Total pressure = $\frac{4}{7} \times 2.1 = 1.2\ \mathrm{atm}$

Q67 Text Solution:

(4)

Europium is f-block element

Q68 Text Solution:

(4)

Heat of hydrogenation$\propto \frac{1}{\text{Stability of alkene}}$
greater the number of α-hydrogen atoms attached to $C=C$ bond greater will be the stability of alkene and minimum will be the heat of hydrogenation of alkene

Q69 Text Solution:

(2)

correct order of dipole moment is $CH_3Cl > CH_3F > CH_3Br$. As electronegativity of halogen atoms decreases from Cl to I.Though F is more eletronegative than Cl, but due to the very small size of F, which outweighs the effect of greater electronegativity

Q70 Text Solution:

(4)

$i = 1-\alpha+n\alpha$
$\alpha = 60\% = 0.6$
$n = 5$ (A_2B_3 gives 5 ions)
$i = 3.4$
$\Delta T_b = i \times K_b \times \text{molality}$
$\Delta T_b = 3.4 \times 0.52 \times 1 = 1.76$
$\therefore$ Boiling point of given solution = 373.15 + 1.7 = 374.76K

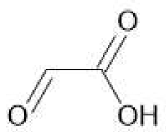
Q71 Text Solution:

(3)

Dissociation of bond will produce 3°-carbocation which is most stable.

Q72 Text Solution:

(2)



IUPAC : Formylmethanoic acid

Q73 Text Solution:

(2)

Resonance is much stronger effect than inductive effect. Thus correct order is

$$CH_2=CH-\overset{+}{C}H_2 > (CH_3)_3\overset{+}{C} > (CH_3)_2\overset{+}{C}H$$

Q74 Text Solution:

(1)

order is correct oreder of -I effect of the substituents is

$-NR_2 < -OR < -F$

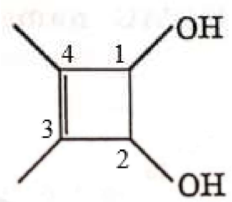
Q75 Text Solution:

(4)

σ^*_{1s}, $\sigma^*_{2p_z}$ and π_{2p_x} have one nodal plane whereas $\pi^*_{2p_y}$ has two nodal planes

Q76 Text Solution:

(4)



IUPAC : 3,4-dimethylcyclobut-3-en-1,2-diol

Q77 Text Solution:

(3)

Molecular weight of $CaCl_2$ = 111 g/mol
111 g of $CaCl_2$ = N_A ions of Ca^{2+}
222 g of $CaCl_2$ = $\frac{N_A \times 222}{111} = 2N_A$ ions of Ca^{2+}
Also, 111 g $CaCl_2$ = $2N_A$ ions of Cl^-
222 g of $CaCl_2$ = $\frac{2N_A \times 222}{111}$ = $4N_A$ ions of Cl^-

Q78 Text Solution:

(3)

for isotonic solution,osmotic pressure should be same

$\pi_{\text{sucrose}} = \pi_{\text{Glucose}}$
$\frac{34.2}{342} \times RT = C \times RT$ or $C = 0.1$
$\therefore$ Concentration of glucose = 0.1×180 = 18 g/L

Q79 Text Solution:

(1)

Less ionisation enthalpy of A means more electropositive nature and high electron affinity means more electronegativity of B. Larger the difference between electronegativity of A and B, more feasible is the formation of ionic bond.

Q80 Text Solution:

(1)

IUPAC : Tetraphenylmethane

Q81 Text Solution:

(3)

$2A + B \rightarrow A_2B$

The product A_2B is a new compound formed hence it does not shows the properties of A and B. The product formed is a compound and not an element.

Q82 Text Solution:

(1)

N^{3-}, O^{2-} and F^{3-} are isoelectronic ions.

Amongst isoelectronic ions, greater the negative charge on the anion, larger is the ionic radius.

Thus, ionic radii decrease in the order :

$$N^{3-}\left(1.71\ \mathring{A}\right) > O^{2-}\left(1.40\ \mathring{A}\right)$$

$$> F^{-}\left(1.36\ \mathring{A}\right)$$

Q83 Text Solution:

(4)

degree of unsaturation in this compound is three.

Q84 Text Solution:

(4)

XeF_2 has three lone pairs and two bond pairs hence hybridisation involved is sp^3d and hence linear whereas H_2O has twon lone pairs and two bond pairs hence hybridisation involved is sp^3 and hence angular in shape.

Q85 Text Solution:

(1)

In octahedral molecule MX_6, six hybrid orbitals (sp^3d^2) directed twoards the corners of a regular octahedron with a bond angle of 90°.

According to this geometry, the number of X – M – X bonds at 180° is three for regular octahedral molecule.

Q86 Text Solution:

(4)

All of these can exhibit optical activity.

Q87 Text Solution:

(2)

Parts per million is defined as the quantity of solute (in grams) present in 10^6 grams of solution,

i.e., $\text{ppm} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6$

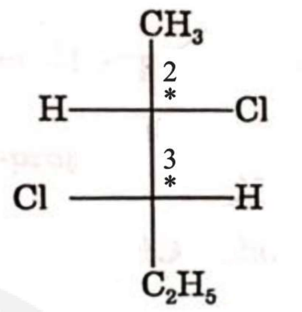
Given, mass of solute = 6 mg = 6×10^{-3} g

mass of solution = 1 kg = 1000 g

$\therefore\ \text{ppm} = \frac{6\times10^{-3}}{1000} \times 10^6 = 6$

Q88 Text Solution:

(2)



The absolute congiguration of the compound is 2S, 3S

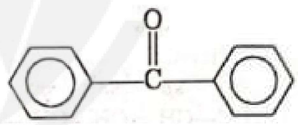
Q89 Text Solution:

(3)

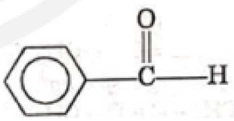
Both assertion and reason are true and reason is the correct explanation of assertion.

Q90 Text Solution:

(4)



is benzophenone



is benzaldehyde

There is no α-hydrogen atom in benzaldehyde and benzophenone hence they do not exhibit tautomerism.

Q91 Text Solution:

(3)

70 g (1 mole) of the liquid has N_A molecules.

$70\ g = \frac{70}{1.2}\text{mL}\ \left(\because\ V = \frac{\text{Mass}}{\text{Density}}\right)$

$\frac{70}{1.2}\text{mL}$ has N_A molecules.

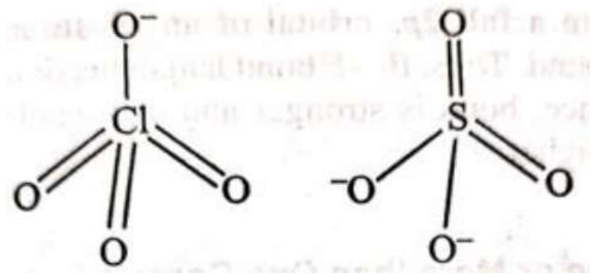
2 mL (or 35 drops) will have

$\frac{N_A}{70} \times 1.2 \times 2 = \frac{1.2\ N_A}{35}$

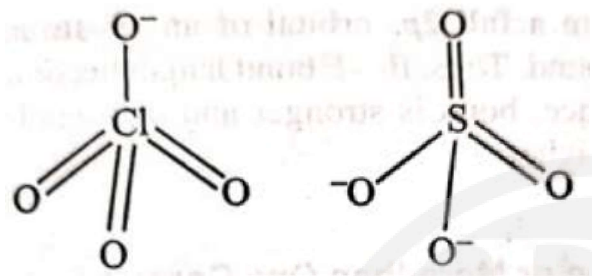
Then 1 drop will have $= \frac{1.2\ N_A}{35\times35} = \frac{1.2\ N_A}{(35)^2}$

Q92 **Text Solution:**

(2)



Bond order : = 1.75 Bond order : = 1.5



Bond order : = 1.25 Bond order : = 1

Greater the bond order, shorter is the bond length.

Hence X–O bond lengths will be in the order:

$SiO_4^{4-} > PO_4^{3-} > SO_4^{2-} > ClO_4^-$

Q93 **Text Solution:**

(4)

Optical isomerism and geometric isomerism

Q94 **Text Solution:**

(1)

Number of geometrical isomers = $2^n = 2^2 = 4$.

Q95 **Text Solution:**

(1)

Elements with atomic number 30, [Ar] $3d^{10}\ 4s^2$

Elements with atomic number 48, [Kr] $4d^{10}\ 5s^2$

Elements with atomic number 80, [Xe] $4f^{14}\ 5d^{10}\ 6s^2$

Q96 **Text Solution:**

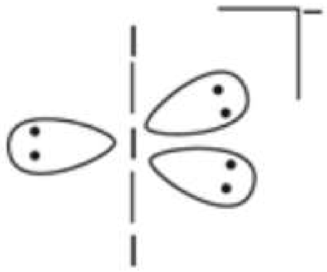
(2)

correct order of stability of carbanion is: a > b > c

Q97 **Text Solution:**

(1)

Hybridisation for $I_3^- = \frac{1}{2}\left[7+2-0+1\right] = 5$



Hybridisation of I in I_3^- is sp^3d

Thus, I_3^- has trigonal bipyramidal geometry with linear shape.

Q98 **Text Solution:**

(3)

More the charge density of cation (i.e., smaller size and high magnitude of charge) more is the covalent character.

Thus the covalent charcter is: $LiCl < BeCl_2 < BCl_3 < CCl_4$

Q99 **Text Solution:**

(3)

H_2SO_4 is 95% by volume

Weight of H_2SO_4 = 95 g

Volume of H_2SO_4 solution = 100 mL

$\therefore$ Moles of $H_2SO_4 = \frac{95}{98}$ and weight of solution = 100×1.98 = 198 g

Weight of water = $198 - 95 = 103$ g

Molality = $\frac{95\times1000}{98\times103} = 9.412$ m

Hence, molality of H_2SO_4 solution is 9.412 m.

Q100 **Text Solution:**

(2)

	Hybridisation	**Lone pair**	**Bond angle**
NH_3	sp^3	1	107°
NH_4^+	sp^3	0	109.5°
NH_2^-	sp^3	2	104°

Thus, correct order is, $NH_4^+ > NH_3 > NH_2^-$

Q101 **Text Solution:**

(4)

The fimbriae are small bristle like fibres sprouting out of the cell. In some bacteria, they are known

to help attach the bacteria to rocks in streams and also to the host tissues.
[New NCERT Class 11th Page No. 91]

Q102 Text Solution:
(2)
Kingdom → Phylum → Class → Order → Family → Genus → Species
As we go higher from species to kingdom, the number of common characteristics goes on decreasing.
[New NCERT Class 11th Page No. 8]

Q103 Text Solution:
(2)
Euglenoids have two flagella.
[New NCERT Class 11th Page No. 15]

Q104 Text Solution:
(3)

- Interphase constitutes more than 95% of the duration of cell cycle.
- M phase is divided into karyokinesis and cytokinesis (Cytoplamic division).

[New NCERT Class 11th Page No. 121]

Q105 Text Solution:
(1)
Roots in some genera have fungal association in the form of mycorrhiza *(Pinus),* while in some others *(Cycas)* small specialised roots called coralloid roots are associated with N_2- fixing cyanobacteria.
[New NCERT Class 11th Page No. 32]

Q106 Text Solution:
(4)
Best stage to study morphology of chromosome is metaphase.
[New NCERT Class 11th Page No. 123]

Q107 Text Solution:
(2)
A. Diplotene is characterised by the presence of chiasmata.
B. In oocytes of some vertebrates, diplotene can last for months or years.
C. Interkinesis is generally short lived.
D. The centriole duplicates during the S-phase of cell cycle in cytoplasm.
[New NCERT Class 11th Page No. 121,126,127]

Q108 Text Solution:
(1)
Saccharomyces is unicellular.
[New NCERT Class 11th Page No. 16]

Q109 Text Solution:
(3)
Zoospore is absent in red algae.
.**[New NCERT Class 11th Page No. 27]**

Q110 Text Solution:
(4)

Diptera	Order
Insecta	Class
Arthropoda	Phylum
Muscidae	Family

[New NCERT Class 11th Page No. 7,8]

Q111 Text Solution:
(4)

- It has 70S type of ribosome present in the stroma.
- Microbodies are present in plants and animals.
- Cillia and flagella have 9+2 arrangement of microtubules.

[New NCERT Class 11th Page No. 94,99,102]

Q112 Text Solution:
(3)
Cell division by mitosis in haploid cell.
[New NCERT Class 11th Page No. 125]

Q113 Text Solution:
(2)
The major lipids are phospholipids that are arranged in a bilayer in plasma membrane.

[New NCERT Class 11th Page No. 93]

Q114 Text Solution:

(3)

I-Spores possess true walls	II-Stiff cellulose plates	III-Gullet

[New NCERT Class 11th Page No. 15]

Q115 Text Solution:

(4)

Natural classification systems developed, were based on natural affinities among the organisms and consider,not only the external features, but also internal features, like ultrastructure, anatomy, embryology and phytochemistry. Such a classification for flowering plants was given by George Bentham and Joseph Dalton Hooker.

[New NCERT Class 11th Page No. 24]

Q116 Text Solution:

(4)

Chemosynthetic autotrophic bacteria oxidise various inorganic substances such as nitrates, nitrites and ammonia and use the released energy for their ATP production. They play a great role in recycling nutrients like nitrogen, phosphorous,iron and sulphur.

[New NCERT Class 11th Page No. 13]

Q117 Text Solution:

(1)

D. Chromatin condensation

C. Synapsis

B. Crossing over

A. Terminalisation

[New NCERT Class 11th Page No. 126]

Q118 Text Solution:

(2)

The most notable disease caused by prions is bovine spongiform encephalopathy (BSE).

Viroids are found to be a free RNA.

[New NCERT Class 11th Page No. 21]

Q119 Text Solution:

(3)

Enzymes of lysosome work in acidic pH.

[New NCERT Class 11th Page No. 96]

Q120 Text Solution:

(1)

The animal order, Carnivora, includes families like Felidae and Canidae.

[New NCERT Class 11th Page No. 5,7]

Q121 Text Solution:

(3)

Plantae	Organ and tissue level of organisation
Monera	Nuclear membrane absent
Fungi	Loose tissue organisation
Protista	Cellular and eukaryotic

[New NCERT Class 11th Page No. 11]

Q122 Text Solution:

(4)

The number of species that are known and described range between 1.7-1.8 million

If we visit a dense forest, we would probably see a much greater number and kinds of living organisms in it.

[New NCERT Class 11th Page No. 3, 4]

Q123 Text Solution:

(1)

Lysosomes-Single membrane

Golgi apparatus-Single membrane

Mitochondria-Double membrane

[New NCERT Class 11th Page No. 95,96]

Q124 Text Solution:

(2)

The sex organs in bryophytes are multicellular. The male sex organ is called antheridium. They produce biflagellate antherozoids.

[New NCERT Class 11th Page No. 29]

Q125 Text Solution:

(1)
Family–Poaceae
[New NCERT Class 11th Page No. 8]

Q126 Text Solution:
(2)
B. Recombination nodule is formed in pachytene.
C. Meiosis involves four cell formation.
D. Parental chromosome replication occur during S phase.
[New NCERT Class 11th Page No. 121,125,126,127,128]

Q127 Text Solution:
(4)
Asexual reproduction takes place by zoospores (motile) or by aplanospores (non-motile). These spores are endogenously produced in sporangium.
[New NCERT Class 11th Page No. 17]

Q128 Text Solution:
(1)
A. Dyad of cells is formed during the telophase-I stage of meiosis
B. Interkinesis is stage during meiosis.
C. Bivalent formation occurs during zygotene.
[New NCERT Class 11th Page No. 126,127]

Q129 Text Solution:
(1)

- False feet is the characteristic of *Amoeboid protozoan.*
- Protozoan do not belong to slime moulds.

[New NCERT Class 11th Page No. 15,16]

Q130 Text Solution:
(3)
Green algae used as a food by space travellers is *Chlorella.*
[New NCERT Class 11th Page No. 26]

Q131 Text Solution:
(4)
The members of chlorophyceae are commonly called green algae. They are usually grass green due to the dominance of pigments chlorophyll *a* and *b.*
[New NCERT Class 11th Page No. 26]

Q132 Text Solution:
(4)
Protist are eukaryotes and it posses plasma membrane and nucleus.
[New NCERT Class 11th Page No. 91]

Q133 Text Solution:
(2)
In plant cells the vacuoles can occupy up to 90 per cent of the volume of the cell.
[New NCERT Class 11th Page No. 96]

Q134 Text Solution:
(2)
Family Poacae is of wheat.
[New NCERT Class 11th Page No. 8]

Q135 Text Solution:
(3)

Feature	**Plant**
Algin	Brown algae
Carrageen	Red algae
Agar	*Gelidium*
Packing material	*Sphagnum*
Prevent soil erosion	Mosses

[New NCERT Class 11th Page No. 24, 29]

Q136 Text Solution:
(3)
Karyogamy take place in the basidium.
[New NCERT Class 11th Page No. 18]

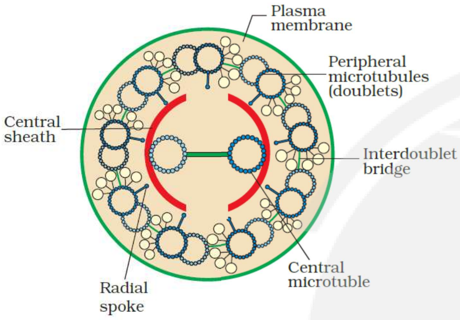
Q137 Text Solution:
(2)
Water moves across the membrane by____osmosis______ while Na^+/K^+ move across plasma membrane by_____active transport______.
[New NCERT Class 11th Page No. 94]

Q138 Text Solution:
(1)

Thylakoids	Flat membranous sacs in stroma
Cristae	Infoldings in mitochondria
Cisternae	Disc-shaped sacs in Golgi apparatus
Chromatin	Condensed structure of DNA

[New NCERT Class 11th Page No. 95,97,98,101]

Q139 Text Solution:
(1)



[New NCERT Class 11th Page No. 99]

Q140 Text Solution:
(3)
Both the words in a biological name, when handwritten, are separately underlined, or printed in italics to indicate their Latin origin.
[New NCERT Class 11th Page No. 4]

Q141 Text Solution:
(3)
Multicellular rhizoids are the feature of *Funaria.*
[New NCERT Class 11th Page No. 29]

Q142 Text Solution:
(3)
The plant body is usually attached to the substratum by a holdfast, and has a stalk, the stipe and leaf like photosynthetic organ – the frond.
[New NCERT Class 11th Page No. 27]

Q143 Text Solution:
(3)
Diatoms are the chief 'producers' in the oceans.
[New NCERT Class 11th Page No. 14]

Q144 Text Solution:
(4)
Thallus of bryophytes is attach to substratum by unicellular and multicellular rhizoids.
Thallus of bryophytes is erect or prostrate.
[New NCERT Class 11th Page No. 29]

Q145 Text Solution:
(1)
A. G_2 – Gap phase 2
B. S – Duplication phase
C. G_1 – Gap phase 1
D. G_0 – Quiescent stage
[New NCERT Class 11th Page No. 121]

Q146 Text Solution:
(3)
They come from both of the individual's parents.
[New NCERT Class 11th Page No. 126]

Q147 Text Solution:
(3)
Viruses did not find a place in classification since they are not considered truly 'living'.
Virus means venom or poisonous fluid.
[New NCERT Class 11th Page No. 19, 20]

Q148 Text Solution:
(3)
Naming is only possible when the organism is described correctly and we know to what organism the name is attached to. This is identification.
The scientific term for the categories is taxa.
[New NCERT Class 11th Page No. 4, 5]

Q149 Text Solution:
(3)
Mycoplasma lack cell wall.
[New NCERT Class 11th Page No. 89]

Q150 Text Solution:
(2)
The common taxonomic categories of mango and wheat are division Angiospermae and Kingdom Plantae.
[New NCERT Class 11th Page No. 7,8]

Q151 Text Solution:

(2)

All these animals belong to phylum Arthropoda, and all arthropods have chitinous exoskeleton.

[New NCERT Class 11th Page No. 44]

Q152 Text Solution:

(3)

The brain in cockroach is located in the head region and is represented by supra-oesophageal ganglion.

[Old NCERT Class 11th Page No. 112]

Q153 Text Solution:

(2)

In frogs, urinary bladder is present ventral to the rectum.

[Old NCERT Class 11th Page No. 112,115,118,119]

Q154 Text Solution:

(2)

The blood leaving the lungs has all its oxygenated haemoglobin and gives up oxygen to the tissues because O_2 concentrated in tissues is low and CO_2 concentration is high in tissues as compared to lungs.

[New NCERT Class 11th Page No. 189]

Q155 Text Solution:

(4)

Every 100 ml of deoxygenated blood delivers approximately 4 ml of CO_2 to the alveoli.

Therefore, 2 litre of deoxygenated blood delivers approximately 80 mL of CO_2 to the alveoli.

[New NCERT Class 11th Page No. 190]

Q156 Text Solution:

(4)

There are four different nitrogenous bases commonly found in DNA; adenine (A), guanine (G), thymine (T) and cytosine (C). RNA also contains adenine, guanine and cytosine but instead of thymine, it has uracil (U). Adenine and guanine are double ring bases called substituted purines. Cytosine, thymine and uracil are single ring bases called substituted pyrimidines. Guanine and cytosine show complementary base pairing and form three hydrogen bonds, while adenine and thymine bind with each other by two hydrogen bonds.

[New NCERT Class 11th Page No. 108]

Q157 Text Solution:

(2)

- Basic amino acids have an additional amino group. For *eg* – Lysine, arginine, etc.
- Neutral amino acids are the ones that possess equal number of amino group and carboxylic group. For *eg*: Valine, glycine, *etc*.
- Serine is an alcoholic amino acid and phenylalanine is an example of aromatic amino acid.

[New NCERT Class 11th Page No. 106]

Q158 Text Solution:

(1)

Respiratory Gas	Atmospheric Air	Alveoli	Blood (Deoxygenated)	Blood (Oxygenated)	Tissues
O_2	159	104	40	95	40
CO_2	0.3	40	45	40	45

[New NCERT Class 11th Page No. 187]

Q159 Text Solution:

(1)

- Antibiotics and coloured pigments are secondary metabolites.
- High temperature destroys the enzymatic activity because proteins are denatured by heat.

[New NCERT Class 11th Page No. 108, 115]

Q160 Text Solution:

(4)

- Collagen is the most abundant protein in the animal world. Tendons and ligaments belong to the category of dense regular connective tissues in which orientation of collagen fibres show a regular pattern.
- Blood is a fluid connective tissue that lacks collagen fibres.

[New NCERT Class 11th Page No. 103, 104]

Q161 Text Solution:

(2)

Catla

Mouth is mostly terminal in fishes of class Osteichthyes.

[New NCERT Class 11th Page No. 119]

Q162 Text Solution:

(1)

Starch is present as a store house of energy in plant tissues. It forms helical secondary structures. In fact, starch can hold I_2 molecules in the helical portion. The starch- I_2 complex is blue in colour. Cellulose does not contain complex helices and hence, cannot hold I_2.

[New NCERT Class 11th Page No. 110]

Q163 Text Solution:

(4)

- Gap junctions facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules.
- Cardiac muscle fibres show densely stained cross-bands called intercalated discs having communication junctions at intervals.
- Cell junctions hold smooth muscle fibres together and they are bundled together in a connective tissue sheath.

[New NCERT Class 11th Page No. 105]

Q164 Text Solution:

(3)

Pigments	Carotenoids, Anthocyanins, *etc.*
Alkaloids	Morphine, Codeine, *etc.*
Terpenoids	Monoterpenes, Diterpenes
Essential oils	Lemon grass oil, *etc.*
Toxins	Abrin, ricin
Lectins	Concanavalin A
Drugs	Vinblastine, curcumin, *etc.*
Polymeric substances	Rubber, gums, cellulose

[New NCERT Class 11th Page No. 108]

Q165 Text Solution:

(2)

The skin of mammals is unique in possessing hair. *Rattus* is a mammal.

[New NCERT Class 11th Page No. 51]

Q166 Text Solution:

(1)

The **correct** order of increasing volume is;
Tidal volume (500 ml) < Expiratory reserve volume (1100 ml) < Inspiratory reserve volume (3000 ml) < Vital capacity (4600 ml)

[New NCERT Class 11th Page No. 187]

Q167 Text Solution:

(3)

- Bones have a hard and non-pliable ground substance, rich in calcium salts and collagen fibres which give bone its strength. The bone cells (osteocytes) are present in the spaces called lacunae.
- The intercellular material of cartilage is solid and pliable and resists compression. Cells of this tissue (chondrocytes) are enclosed in small cavities within the matrix secreted by them.

[Old NCERT Class 11th Page No. 104]

Q168 Text Solution:

(1)

- The long protein chain or the polypeptide chain usually folds upon itself like a hollow woolen ball. This is termed as the tertiary structure. This structure gives a 3-dimensional view of a protein.
- Tertiary structure is absolutely necessary for the many biological activities of proteins.
- Hydrogen bond, ionic bond, covalent bond, hydrophobic bond as well as van der Waals

interaction are involved during coiling of a polypeptide.

[New NCERT Class 11th Page No. 112]

Q169 Text Solution:

(4)

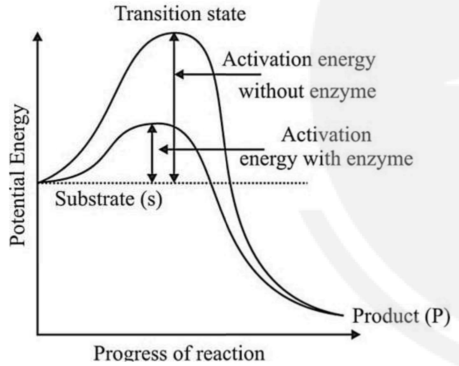
- Neural tissue exerts the greatest control over the body's responsiveness to changing conditions.
- Neurons, the functional units of neural system are excitable cells due to a differential concentration gradient of ions across the membrane.
- The neuroglial cells which constitute the rest of the neural system protect and support neurons.

[New NCERT Class 11th Page No. 105]

Q170 Text Solution:

(2)

The graph represents an exothermic reaction.



[New NCERT Class 11th Page No. 115]

Q171 Text Solution:

(3)

Felis are homeotherms because it can maintain a constant body temperature.

[New NCERT Class 11th Page No. 51]

Q172 Text Solution:

(3)

Hydrolases catalyse the hydrolysis of ester, ether, peptide, glycosidic , C-C, C-halide or P–N bonds. Hydrolases belong to class III of enzymes.

[New NCERT Class 11th Page No. 117]

Q173 Text Solution:

(1)

Simple squamous epithelium	–	Forms diffusion boundary
Simple cuboidal epithelium	–	Assists in secretion and absorption in the proximal convoluted tubule of kidney
Ciliated epithelium	–	Moves mucus in a specific direction
Compound epithelium	–	Provides protection against chemical and mechanical stresses

[New NCERT Class 11th Page No. 101]

Q174 Text Solution:

(3)

- Inulin, glycogen and chitin are homopolysaccharides while GLUT-4 and antibody belong to the category of proteins.
- GLUT-4 enables glucose transport into the cells while antibody fights infectious agents.

[New NCERT Class 11th Page No. 109]

Q175 Text Solution:

(1)

- Connective tissues are the most abundant and widely distributed tissue in the body of complex animals. They are named connective tissues because of the special function of linking and supporting other tissues/organs of the body. They range from soft connective tissues to specialised types.
- Muscles play an active role in all the movements of the body.
- Epithelial tissues line the body cavities, ducts and tubes.

[New NCERT Class 11th Page No. 103]

Q176 Text Solution:

(3)

The tracheae, primary, secondary and tertiary bronchi, and initial bronchioles are supported by incomplete cartilaginous rings.

[New NCERT Class 11th Page No. 184]

Q177 Text Solution:

(4)

- The wall of internal organs such as the blood vessels, stomach and intestine contains smooth muscle tissues.
- Skeletal muscle fibres are present in the limbs, body wall, face, neck, *etc.*
- Skeletal as well as cardiac muscle fibres are striated and cylindrical in shape.
- Skeletal as well as smooth muscle fibres do not possess intercalated discs.
- Both smooth as well as cardiac muscle fibres are involuntary in action.

[New NCERT Class 11th Page No. 105]

Q178 Text Solution:

(3)

- Carbohydrates constitute 3% of the total cellular mass. In a polysaccharide, the individual monosaccharides are linked by glycosidic bonds. Cotton fibres are cellulosic and its monomeric units *e.*, glucose, are joined together by glycosidic bonds.
- In nucleosides, the nitrogenous base is joined to the sugar molecule by a glycosidic bond. RuBisCO are the most abundant protein in the whole of the biosphere. The monomeric units of proteins are joined by peptide bonds.

[New NCERT Class 11th Page No. 109]

Q179 Text Solution:

(1)

- Sulphate and phosphate are present in the acid soluble fraction of living tissue.
- Arachidonic acid has 20 carbon atoms including the carboxyl carbon.

[New NCERT Class 11th Page No. 105,115]

Q180 Text Solution:

(3)

Hepatic caeca in cockroaches is a part of the digestive system, 6-8 in number and is present at the junction of foregut and midgut and it secretes digestive juices.

[New NCERT Class 11th Page No. 113]

Q181 Text Solution:

(3)

Platyhelminthes (*Taenia*) exhibit bilateral symmetry.

[New NCERT Class 11th Page No. 43]

Q182 Text Solution:

(4)

In the alveoli, where there is high pO_2, low pCO_2, lesser H^+ concentration and lower temperature, the factors are all favourable for the formation of oxyhaemoglobin, whereas in the tissues, where low pO_2, high pCO_2, high H^+ concentration and higher temperature exist, the conditions are favourable for dissociation of oxygen from the oxyhaemoglobin.

[New NCERT Class 11th Page No. 189]

Q183 Text Solution:

(3)

Mammary glands	Platypus
Proboscis gland	*Saccoglossus*
Alternation of generation	*Obelia*
Bioluminescence	*Ctenoplana*

[New NCERT Class 11th Page No. 40-51]

Q184 Text Solution:

(2)

- Prosthetic groups are organic compounds and are distinguished from other co-factors in that they are tightly bound to the apoenzyme. For example, in peroxidase and catalase, which catalyze the breakdown of hydrogen peroxide to water and oxygen, haem is the prosthetic group and it is a part of the active site of the enzyme.
- Zn^{2+} is the co-factor for the enzymes, carboxypeptidase and carbonic anhydrase.

[New NCERT Class 11th Page No. 118]

Q185 Text Solution:
(2)

- Primary metabolites have identifiable functions and play known roles in the physiological processes.
- Cellulose is a secondary metabolite. Glucagon and insulin are hormones that are proteinaecous in nature.
- Inulin is a polymer of fructose.

[New NCERT Class 11th Page No. 108]

Q186 Text Solution:
(1)

- In triglyceride, three alkyl group are present.
- Two hydrogen bonds are present between adenine and thymine.
- Lecithin is a phospholipid which is composed of two fatty acids, a phosphate group and a nitrogen containing choline molecule.
- Glycine is the simplest amino acid.
- Two hydrogen atoms are directly attached to the alpha C atom in glycine.

[New NCERT Class 11th Page No. 107]

Q187 Text Solution:
(1)
Pleural fluid between the double layered pleura Membrane of the lungs reduces friction on the lung-surface.
[New NCERT Class 11th Page No. 184]

Q188 Text Solution:
(4)

- *Petromyzon* and *Myxine* are examples of class Cyclostomata.
- *Pristis* is an example of class Chondrichthyes.
- *Corvus* and *Columba* both are examples of class Aves which possess pneumatic bones.

[New NCERT Class 11th Page No. 47]

Q189 Text Solution:
(2)
Trachea is a straight tube which divides at the level of 5th thoracic vertebra.
[New NCERT Class 11th Page No. 187]

Q190 Text Solution:
(3)

- The cuboidal epithelium is composed of a single layer of cube-like cells. This is commonly found in ducts of glands and tubular parts of nephrons in kidneys and its main function is secretion and absorption.
- Tendons attach skeletal muscles to bones while ligaments attach one bone to another.
- Neurons possess axon, dendrites as well as cyton but not neuroglia.

[New NCERT Class 11th Page No. 102/103,104]

Q191 Text Solution:
(1)

- In inspiration, diaphragm becomes flat shaped.
- In inspiration, contraction of diaphragm muscle increases the volume of thoracic cavity.

[New NCERT Class 11th Page No. 186]

Q192 Text Solution:
(3)
The sense organs in cockroaches are eyes, antennae, labial palps, maxillary palps and anal cerci.
[New NCERT Class 11th Page No. 114]

Q193 Text Solution:
(4)

- The body cavity which is lined by mesoderm is present in
- Notochord is a rod-like structure formed on the dorsal side during embryonic development in
- *Euspongia* exhibit cellular level of organisation.
- *Neophron* and *Macaca* are warm blooded animals.

[New NCERT Class 11th Page No. 39, 40, 47]

Q194 Text Solution:
(2)

When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibition. Due to its close structural similarity with the substrate, the inhibitor competes with the substrate for the substrate binding site of the enzyme. Consequently, the substrate cannot bind and as a result, the enzyme action declines.

[New NCERT Class 11th Page No. 117]

Q195 Text Solution:

(4)

Male frogs can be distinguished from female frogs by the presence of sound producing vocal sacs and presence of copulatory pads on the first digits of forelimbs.

[New NCERT Class 11th Page No. 116]

Q196 Text Solution:

(4)

- Chitin - Exoskeleton of arthropods
- Collagen - Intercellular ground substance
- Anthocyanins - Pigments
- Coenzyme NADP- Vitamin Niacin

[New NCERT Class 11th Page No. 108,112,118]

Q197 Text Solution:

(1)

- Amphibians, reptiles, birds and mammals respire through lungs.
- Amphibians like frogs can respire through their moist skin (cutaneous respiration) also.

[New NCERT Class 11th Page No. 183]

Q198 Text Solution:

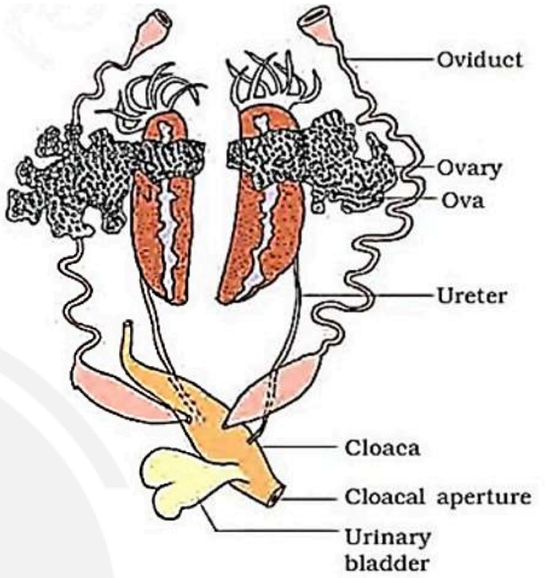
(3)

- In Urochordata(*Doliolum*), notochord is present only in larval tail.
- Cyclostomes (*Myxine*) are marine but migrate for spawning to fresh water.

[New NCERT Class 11th Page No. 46, 47, 49]

Q199 Text Solution:

(4)



- The diagram depicts the female reproductive system of frogs.
- Urinary bladder stores urine until it is excreted out.

[New NCERT Class 11th Page No. 48]

Q200 Text Solution:

(4)

Scoliodon	Gills
Delphinius	Lungs
Hydra	Entire body surface
Insects	Tracheal tubes

[New NCERT Class 11th Page No. 184]